

How Literacy and Age at Marriage Shape Family Size in Portugal, 1979*

A Statistic Research Using Poisson Regression Models

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TBD. IDK. Smash Them

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*Code and data are available at: https://github.com/Jie-jiao05/Family_Size.

1 Introduction

1.1 Why this topic is important

1.2 3 citation relate with our research

1.3 Our objective, goal, how the citation related to, what we gonna do

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2 Method

2.1 Use GLM, possion, (reason), what is x how to interoperate

3 Result

3.1 Data Description

This dataset originates from the 1979-80 Portuguese Fertility Survey (Instituto Nacional de Estatística (Portugal) 1979), conducted as part of the World Fertility Survey (WFS) program and archived by Princeton University's Office of Population Research (OPR).

The Dataset includes 5,148 observations and 8 variables, covering factors such as literacy, marriage age, pregnancies, children, and regional demographics. Table 1 it shows a summary information of our dataset.

To analyze the relationship between family size, literacy, and age at marriage, the variable Literacy was categorized as a dummy variable, assigning values 0 and 1 to illiterate and literate individuals, respectively. The analysis primarily focused on demographic factors, economic influences, and regional differences, aiming to assess their impact on family size. Additionally, an offset variable accounting for the duration of marriage was included to control for varying exposure times.

3.2 Model Selection

Our Final model is showing above:

Table 1: Dataset Summary

X	age	ageMarried	monthsSinceM
Min. : 1	Min. :15.00	Length:5148	Min. : 0.0
1st Qu.:1288	1st Qu.:28.75	Class :character	1st Qu.: 73.0
Median :2574	Median :36.00	Mode :character	Median :148.0
Mean :2574	Mean :35.30		Mean :155.2
3rd Qu.:3861	3rd Qu.:43.00		3rd Qu.:230.0
Max. :5148	Max. :49.00		Max. :431.0
pregnancies	children	sons	literacy
Min. : 0.000	Min. : 0.00	Min. : 0.000	Min. :0.0000
1st Qu.: 1.000	1st Qu.: 1.00	1st Qu.: 0.000	1st Qu.:1.0000
Median : 2.000	Median : 2.00	Median : 1.000	Median :1.0000
Mean : 2.282	Mean : 2.26	Mean : 1.187	Mean :0.8871
3rd Qu.: 3.000	3rd Qu.: 3.00	3rd Qu.: 2.000	3rd Qu.:1.0000
Max. :16.000	Max. :17.00	Max. :10.000	Max. :1.0000
region10.20k	region20k.	regionlisbon	regionlt10k
Min. :0.00000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.00000	Median :0.0000	Median :0.0000	Median :1.0000
Mean :0.08411	Mean :0.1132	Mean :0.0913	Mean :0.6803
3rd Qu.:0.00000	3rd Qu.:0.0000	3rd Qu.:0.0000	3rd Qu.:1.0000
Max. :1.00000	Max. :1.0000	Max. :1.0000	Max. :1.0000
regionporto			
Min. :0.00000			
1st Qu.:0.00000			
Median :0.00000			
Mean :0.03108			
3rd Qu.:0.00000			
Max. :1.00000			

$$\text{children}_i \sim \text{Poisson}(\lambda_i) \quad (1)$$

$$\log(\lambda_i) = -4.13839 - 0.12906 \cdot \text{literacy}_i \quad (2)$$

$$+ 0.03536 \cdot \text{ageMarried15to18}_i + 0.02006 \cdot \text{ageMarried18to20}_i \quad (3)$$

$$- 0.01574 \cdot \text{ageMarried20to22}_i - 0.03959 \cdot \text{ageMarried22to25}_i \quad (4)$$

$$- 0.02093 \cdot \text{ageMarried25to30}_i - 0.01054 \cdot \text{ageMarried30toInf}_i \quad (5)$$

$$- 0.06783 \cdot \text{region10_20k}_i - 0.20067 \cdot \text{region20k_}_i \quad (6)$$

$$- 0.18386 \cdot \text{regionlisbon}_i + 0.09381 \cdot \text{regionlt10k}_i \quad (7)$$

$$+ \text{NA} \cdot \text{regionporto}_i + \log(\text{monthsSinceM}_i) \quad (8)$$

3.3 Model selection (use model with offset)

3.4 Use likelyhood rario test to do the performance , to see the efficiency between model

3.5 Test overdispersion

4 Conclusion

Appendix

References

Instituto Nacional de Estatística (Portugal). 1979. *Portugal World Fertility Survey (WFS), 1979-80*. World Fertility Survey (WFS), Demographic; Health Surveys (DHS) Program.
<https://wfs.dhsprogram.com/index.cfm?ccode=pt>.